

A BBP-style computation for π in base 10

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ABSTRACT. We articulate about how to compute (promptly) the digits of π , in bases 5 and 10, from a given place without computing preceding ones.

1. THE BBP ALGORITHM

Given a base $b \in \mathbb{Z}_{\geq 2}$ and a ‘related to the base’ series expansion of constant $\xi \in \mathbb{R}$ (a BBP-type formula), the Bailey–Borwein–Plouffe algorithm [1] computes r base b digits of ξ beginning at the position $d + 1$ after the floating point. A traditional illustrative example is the constant

$$\log 2 = \sum_{n=1}^{\infty} \frac{1}{n2^n}$$

and $b = 2$. To access what begins at the binary position $d + 1$, we look for the fractional part of

$$2^d \log 2 = \sum_{n=1}^d \frac{2^{d-n}}{n} + \sum_{n=d+1}^{\infty} \frac{2^{d-n}}{n}.$$

In the first (finite) sum fractional parts of individual terms are computed using

$$\frac{2^{d-n}}{n} \bmod \mathbb{Z} = \frac{2^{d-n} \bmod n}{n},$$

with $2^{d-n} \bmod n$ calculated with the help of fast modular exponentiation; the second sum — the tail — converges quickly at rate $1/2$, so that its first few digits can be easily computed.

On the other hand, one can vary this recipe slightly by writing

$$2^d \log 2 = \sum_{n=1}^d \frac{2^{d-n}}{n} + \sum_{n=d+1}^{d+2r} \frac{2^{d-n}}{n} + \sum_{n=d+2r+1}^{\infty} \frac{2^{d-n}}{n}.$$

For the first sum here we do exactly the same as before — compute individual fractional parts with the help of fast exponentiation. For the second sum we just sum up the corresponding $2r$ fractions, each being the reciprocal of an integer, while for the third sum we use the estimate

$$\left| \sum_{n=d+2r+1}^{\infty} \frac{2^{d-n}}{n} \right| < \frac{1}{d+2r+1} \sum_{n=d+2r+1}^{\infty} 2^{d-n} = \frac{2^{-2r}}{d+2r+1}$$

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showing that this tail will not in general affect the first r binary digits of $\{2^d \log 2\}$.

2. A BBP-TYPE FORMULA FOR π

Several BBP-type formulae for π are known for bases which are powers of 2. This gives access, for example, to computing hexadecimal digits of π starting at a particular far-away position without computing predecessors. It is an open question whether this type of computation is possible for π related to other bases.

In what follows i stands for $\sqrt{-1}$.

Rational approximations to π constructed by Salikhov in [2, 3] and later refined by Zeilberger and this author in [4] are based on the representation

$$\sum_{n=0}^{\infty} \frac{1}{2n+1} \left(\frac{1}{1-2i} \right)^{2n+1} - \sum_{n=0}^{\infty} \frac{1}{2n+1} \left(\frac{1}{1+2i} \right)^{2n+1} = \frac{\pi i}{4}. \quad (1)$$

Because the norm of the quadratic irrationalities $1 \pm 2i$ is 5, one can think of the expansion as of ‘base $\sqrt{5}$ ’ BBP-type formula for π — in fact ‘base 5’ as we encounter the powers of $(1 \pm 2i)^2$ rather than of $1 \pm 2i$. Can it be used for computing base 5 digits of π ?

Before proceeding with this, we make some related comments about computation of powers $(1 \pm 2i)^n$. Their real and imaginary parts are read off from the 2×2 matrix $\begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}^n$; they are also generated via the recurrence equation $a_n = 2a_{n-1} - 5a_{n-2}$. The latter circumstance leads to the following interpretation of identity (1): Define the sequence b_n through the recursion $b_n = -6b_{n-1} - 25b_{n-2}$ for $n \geq 2$ and initial data $b_0 = 1$, $b_1 = -1$. Then

$$\sum_{n=0}^{\infty} \frac{1}{2n+1} \frac{b_n}{5^{2n}} = \frac{5\pi}{16}$$

and $|b_n| < 2 \cdot 5^n$ for all n .

Formula (1) allows one to cast π as the imaginary part of

$$\xi = 8 \sum_{n=0}^{\infty} \frac{1}{2n+1} \left(\frac{1}{1-2i} \right)^{2n+1},$$

and we can use the BBP strategy to compute the base 5 expansion for both its real and imaginary parts beginning at the position $d+1$ till the position $d+r$, say, under a mild condition we give below (see condition (2)). This demands for computing first r base 5 digits after the floating point of the number $5^d \xi$ (again we do this for both real and imaginary parts!). Because $|1-2i| = \sqrt{5}$, the tail

$$8 \cdot 5^d \sum_{n=d+2r}^{\infty} \frac{1}{2n+1} \left(\frac{1}{1-2i} \right)^{2n+1}$$

is bounded above by

$$\frac{8 \cdot 5^d}{2d+4r+1} \sum_{n=d+2r}^{\infty} \frac{1}{5^{n+1/2}} = \frac{2 \cdot 5^{1/2-2r}}{2d+4r+1},$$

which means that the positions before the position $r + 1$ in the base 5 expansion of $5^d \xi$ are hardly affected, and we look for the first r digits of

$$\sum_{n=0}^{d+2r-1} \frac{8 \cdot 5^d}{2n+1} \left(\frac{1}{1-2i} \right)^{2n+1}.$$

We are only interested in this expression in $\mathbb{Q}[i]$ modulo 1 for real and imaginary parts. Assuming that

$$\frac{\log(2d+4r-1)}{\log 5} \leq d \quad (2)$$

and writing $2n+1 = 5^k \cdot m$ with $(m, 5) = 1$ for the n th term in the sum, we look for

$$\frac{8 \cdot 5^d}{2n+1} \left(\frac{1}{1-2i} \right)^{2n+1} \bmod \mathbb{Z}[i] = \frac{8 \cdot 5^{d-k} (1-2i)^{-(2n+1)} \bmod m}{m};$$

thus, we only need executing fast exponentiation for $5^{d-k} \bmod m$ and $(1-2i)^{2n+1} \bmod m$ in $\mathbb{Z}[i]$, and then computing the reciprocal of the latter modulo m . The latter inversion is possible because m is coprime with the norm of any power of $1-2i$; alternatively, one can take an integer a such that $a \equiv \frac{1}{5} \bmod m$, with the motive that $(1-2i)^{-1} = \frac{1}{5} + \frac{2}{5}i \equiv a(1+2i) \bmod m\mathbb{Z}[i]$, and compute $a^{2n+1}(1+2i)^{2n+1} \bmod m$.

Surprisingly enough, we can use the same scheme for computing *decimal* digits of π from position $d+1$ to $d+r$ by ignoring the tail

$$8 \cdot 10^d \sum_{n=2d+2r}^{\infty} \frac{1}{2n+1} \left(\frac{1}{1-2i} \right)^{2n+1}$$

and summing up, now for n from 0 to $2d+2r-1$, the fractional parts

$$\frac{8 \cdot 10^d}{2n+1} \left(\frac{1}{1-2i} \right)^{2n+1} \bmod 1 = \frac{8 \cdot 2^d \cdot 5^{d-k} ((1-2i)^{2n+1})^{-1} \bmod m}{m}.$$

A different strategy, which is based on the computation of Bernoulli numbers and does not go beyond $d = 10^8$ at the moment of writing, is proposed by Plouffe in [5].

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